

Assignment – 2

Course: Interactive Visual Analysis of Weather Prediction Data During an Aircraft Campaign
for Weather Research

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1. Jetstream Core Lines with Clouds and Wind Speed Isosurface (3D)

At the selected valid time **2025-12-11T09:00:00Z**, the three-dimensional visualization clearly highlights a strong upper-level jet stream located near the **300 hPa pressure level**. The maximum wind speeds are concentrated within the jet core, which appears as a long, curved, tube-like structure. This geometry illustrates both the horizontal extent and vertical depth of the jet stream, showing that it bends across the domain rather than following a straight path.

Cloud formations are mainly aligned along the jet stream axis. Regions with colder cloud tops are positioned near areas of intensified wind speeds, suggesting a close relationship between upper-level wind dynamics and cloud development. In contrast, areas farther away from the jet core display weaker wind speeds and comparatively less cloud coverage.

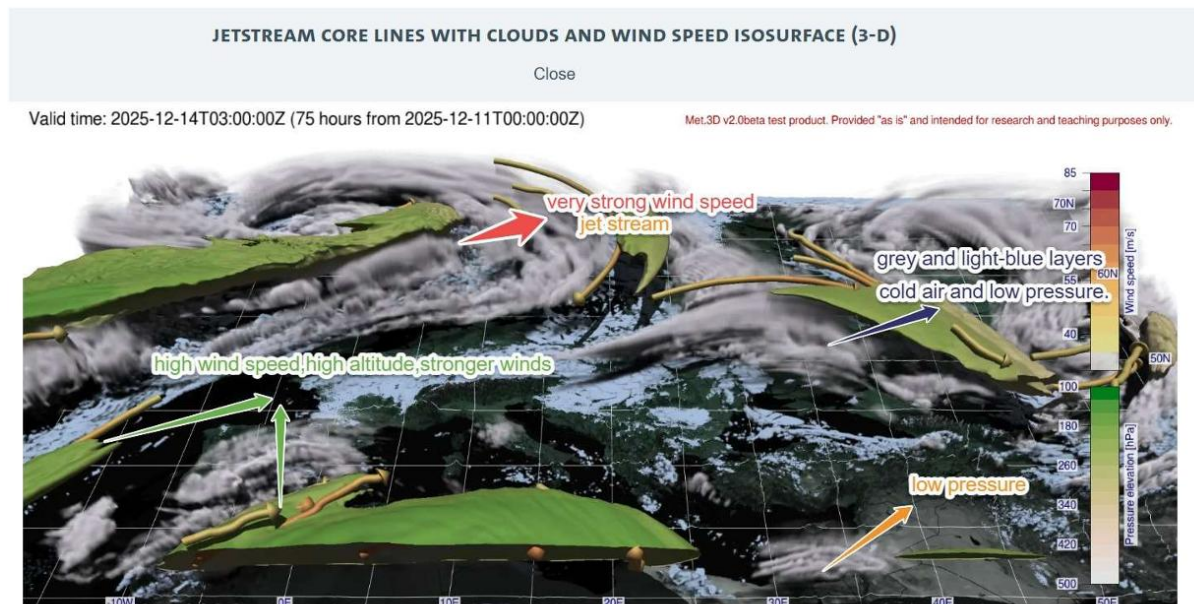
The wind-speed distribution is consistent with the pressure-gradient force. Relatively higher pressure gradients at upper levels correspond to stronger winds, while weaker gradients are associated with slower flow. The alignment of clouds along the jet path indicates that upper-tropospheric winds strongly influence cloud transport and organization.

The interactive 3D visualization allows adjustment of wind-speed thresholds, viewing angles, and vertical levels. This interactivity enhances understanding of jet–cloud interaction, wind-speed gradients, and the spatial structure of low-pressure systems. Overall, the 3D representation effectively demonstrates how upper-level atmospheric processes contribute to large-scale weather system evolution.

Further Investigation

Although the 3D visualization provides valuable spatial insight, it does not fully reveal how wind speed varies over topography or subtle vertical variations within atmospheric layers. Additional analysis would be useful to understand:

- The temporal variability of the jet stream.
- How fluctuations in jet intensity influence surface weather conditions.



2. Jetstream Core Lines with Clouds Coloured by Temperature (2D)

At the selected time step in December 2025, the two-dimensional visualization presents jet core lines together with wind speed and cloud-top temperature fields. The jet stream appears as a continuous, curved band of enhanced wind speeds extending across the region.

Areas characterized by low cloud-top temperatures (indicating colder and deeper cloud systems) generally coincide with regions of stronger upper-level winds. Conversely, regions with relatively warmer cloud tops are associated with weaker wind speeds. This spatial relationship suggests that strong upper-level flow supports the development and maintenance of deep cloud systems.

The cloud structures closely follow the curvature of the jet stream, demonstrating that the wind field controls their horizontal displacement. Regions with strong wind vectors are typically located near colder cloud zones, while warmer cloud regions often show limited wind activity.

Surface wind streamlines further illustrate how the large-scale circulation influences near-ground atmospheric motion, particularly around low-pressure centers. The tightly packed pressure contours around the storm system indicate a well-developed cyclonic structure, typical of a mid-latitude storm system rather than a tropical cyclone.

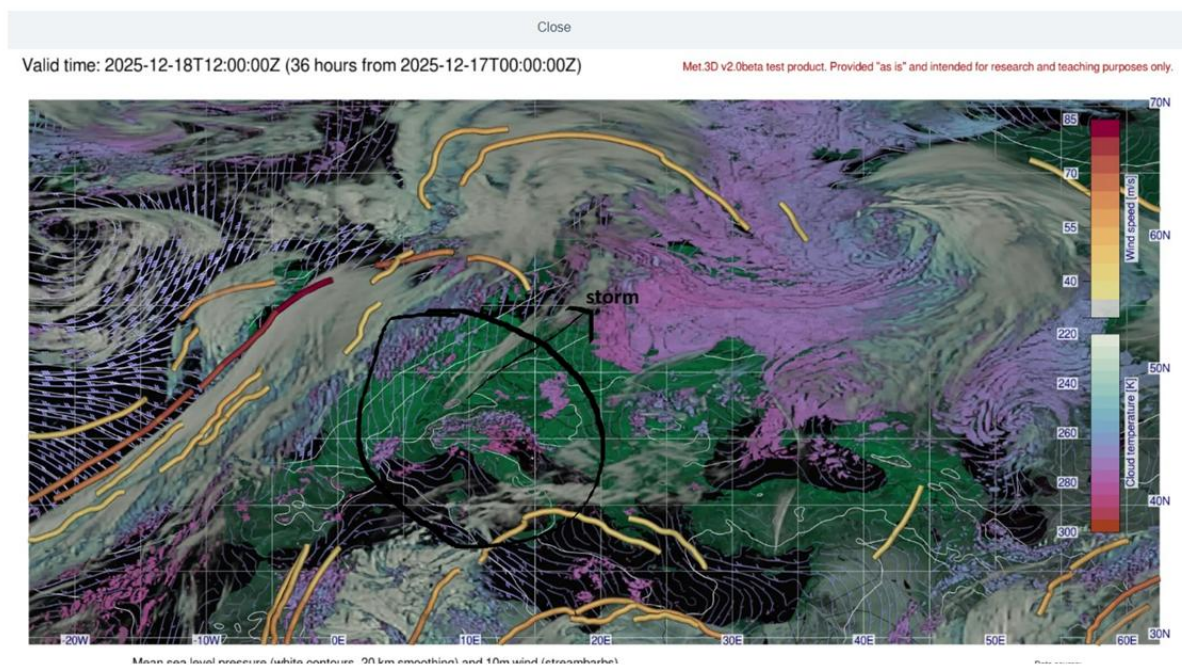
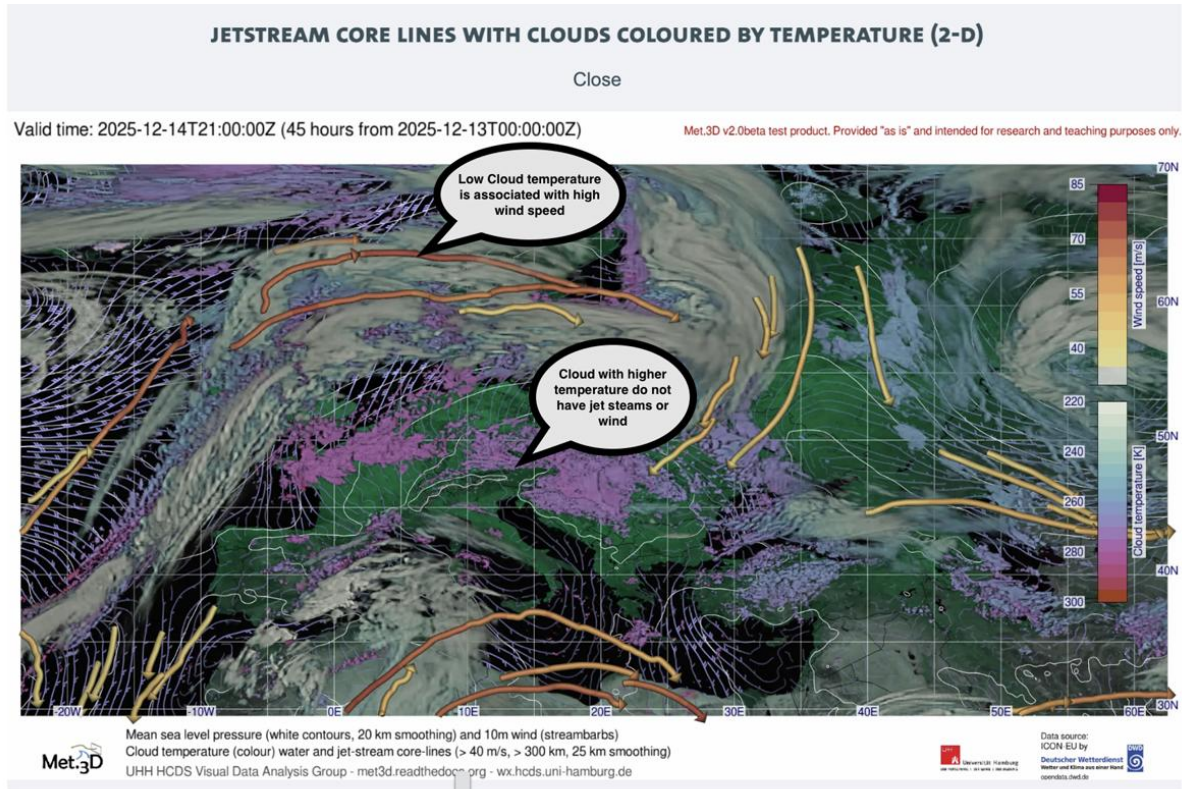
Further Investigation

For deeper understanding of storm dynamics, the following aspects could be examined:

- The strength of the storm at the surface level.
- Whether the system produces heavy rainfall, snowfall, or convective activity.

- The physical mechanisms responsible for strengthening or weakening of the jet stream over time.
- The relationship between humidity, cloud density, temperature, and wind speed.

The interactive Met.3D environment allows modification of time steps and parameter thresholds, enabling detailed examination of how jet streams, cloud systems, and pressure patterns evolve dynamically.



3. Warm Conveyor Belt (WCB) Trajectories (3D)

The 3D trajectory visualization of the Warm Conveyor Belt (WCB) spans the period from **17th to 21st December 2025**, illustrating the temporal evolution of ascending air parcels within the synoptic-scale circulation.

The trajectories show upward motion from higher-pressure levels toward lower-pressure levels, which is consistent with ascent in pressure coordinates. As time progresses, the air parcels also exhibit horizontal displacement, particularly westward movement, indicating that vertical ascent occurs simultaneously with horizontal advection along the large-scale flow.

During the earlier phase, most trajectories are concentrated in the northern part of the domain and follow curved paths around multiple cyclonic centers. These spiral-like trajectories reflect the rotational structure of the atmospheric flow rather than straight vertical ascent.

Around 19 December, a noticeable shift occurs in the spatial distribution of trajectories. The number of ascending parcels decreases in the northern region and becomes more concentrated toward the southern part of the domain. This change suggests a modification in dominant flow patterns and indicates that WCB activity either weakens or relocates during the later stage of the analysis period.

